

Thermodynamics

BRANCH: MECHANICAL ENGG. (ME)

COURSE CODE: ME201

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Thermodynamics (ME201) under the Mechanical Engg. (ME) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Basic Concepts & Zeroth Law

Macroscopic vs Microscopic viewpoint, thermodynamic system, state, processes, cycles. Zeroth Law: Temperature measurement, scale calibration.

Unit 2: First Law of Thermodynamics

First Law applied to a closed system undergoing a process/cycle. Steady Flow Energy Equation (SFEE) applied to nozzles, turbines, compressors, boilers, heat exchangers.

Unit 3: Second Law of Thermodynamics

Limitations of First Law. Heat engines, refrigerators, heat pumps. Kelvin-Planck and Clausius statements. Carnot cycle, Clausius inequality, entropy calculation.

Unit 4: Pure Substances & Cycles

Properties of pure substances, dryness fraction, T-s and h-s diagrams. Vapor power cycles: Rankine cycle. Gas power cycles: Otto, Diesel, and Dual cycles.

First Law: $Q - W = \Delta U$, Entropy: $dS = dQ / T$. COP of Heat Pump = COP of Refrigerator + 1.

CRITICAL FORMULA / PRINCIPLE:

Key Formulas & Exam Pointers



